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Dialogic information retrieval and the falsifiability of Wilson's model: towards a framework for AI-mediated information behaviour

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Abstract

Introduction. This paper revisits Tom D. Wilson's model of information behaviour through the lenses of falsifiability, mechanistic explanation, and dialogic interaction in AI mediated environments. While classical LIS models, such as Bates's berrypicking and Marchionini's exploratory search emphasise iterative and adaptive strategies, Wilson's framework remains difficult to test empirically, because its central constructs are broad and operationally under specified.

Method. The paper adopts a conceptual methodological approach that brings together Popper's and Lakatos's accounts of scientific demarcation with Cronbach and Meehl's framework of construct validity. It develops the Dialogic Mechanistic Information Behaviour Framework as a proposal for translating selected elements of Wilson's model into more explicitly testable propositions.

Analysis. The paper combines theoretical reconstruction and analytical mapping with a small set of illustrative human AI interaction logs used solely to demonstrate how candidate mediators and observable indicators may be interpreted in context.

Results. The framework distinguishes conceptual, mechanistic, and observational levels and clarifies how selected elements of Wilson's model may be linked to candidate mediators and trace level indicators in AI mediated dialogue.

Conclusion. Wilson's model may be reconsidered as a macro level framework whose empirical vulnerability becomes clearer through mechanistic and operational specification in trace rich AI mediated settings.

Introduction

Research on information behaviour has evolved through several major models that describe how individuals recognise, search for, and use information. One of the enduring unresolved issues in this tradition concerns the limited empirical testability of many influential frameworks, especially when they are used as broad conceptual architectures rather than as sources of explicit and disconfirmable propositions. This paper addresses that problem in relation to dialogic, AI-mediated information retrieval and introduces the Dialogic-Mechanistic Information Behaviour Framework (DMIBF) as a conceptual-methodological response.

The problem of non-falsifiability in information behaviour theory

For over four decades, information behaviour research has been shaped by models explaining how individuals recognise, seek, and use information. Frameworks such as Marcia Bates's berrypicking model (1989) and Gary Marchionini's exploratory search (1995) introduced dynamic, iterative views of information seeking, highlighting continual reformulation of needs and queries rather than linear progression. While these models captured the adaptive and evolving nature of searching with considerable analytical force, they remained primarily heuristic and descriptive, offering only limited possibilities for direct empirical refutation.

In parallel, Wilson's models of information behaviour (1981, 1996, 1999) generalised these insights into a macro-level framework linking information need, context, barriers, motivation, and use. Whereas Bates and Marchionini focused on how people search, Wilson addressed why such behaviour occurs and which systemic factors shape it. Although Wilson's framework established a durable conceptual vocabulary for adaptive and context sensitive information behaviour, its breadth and integrative scope have also made empirical operationalisation difficult, particularly in relation to explicit predictive claims.

Falsifiability is a central epistemological criterion for theory development. Popper (1959; 2014) defined it as openness to empirical disconfirmation, while Lakatos (1970) reframed the issue in terms of research programmes, in which relatively stable theoretical commitments are accompanied by revisable auxiliary hypotheses (Godfrey Smith, 2003). Reconsidering Wilson's model from this perspective does not imply that the model has failed as theory. Rather, it raises the question of under what conditions selected elements of the model may be reformulated in more explicitly testable terms.

Although often treated as binary, falsifiability is frequently understood as gradable. In information science, the blurred distinction between models and theories (Bates, 2005) complicates the formulation of clear criteria for testability. Information behaviour research includes many models as simplified representations of reality (Wilson, 2016; Greifeneder & Schlebbe, 2022), differing in scope and testability from basic models to macro- and meta-models. Wilson's classical model is usually treated as a macro-model, though later versions include meta-model elements. For this reason, discussing the falsifiability of a model requires some caution, since models often function less as directly testable propositions than as structures from which testable hypotheses may be derived. Classical information behaviour models typically outline relationships without explicit hypotheses (Garg, 2016; Wilson, 2016), a long-standing feature of the field (Savolainen, 2021; Ingwersen & Järvelin, 2005). Empirical vulnerability therefore arises not at the level of abstract formulation alone, but through the specification of mediator indicator relations, scope conditions, and potential disconfirming observations. In practice, however, information science has more often favoured model revision or extension than explicit disconfirmation. This pattern is evident in successive revisions of Wilson's models, as well as in related frameworks proposed by Niedźwiedzka (2003) and Bawden & Robinson (2022).

Recent reviews in information science note that much theorising in the field has prioritised inductivist extension and positive confirmation over refutation, with the consequence that classical models tend to be descriptively rich yet empirically invulnerable (Ekeland, Galichon & Henry, 2010; Salovaara & Merikivi, 2015). The present paper responds to these concerns by outlining conditions under which selected elements of such models may be translated into more explicitly testable propositions in dialogic, AI mediated settings.

From dialogic retrieval to mechanistic falsification

Contemporary information behaviour is being rapidly reshaped by artificial intelligence, particularly large language models (LLMs) such as ChatGPT. These systems enable a dialogic form of information retrieval based on iterative prompting, reflection, and evaluation, in which users repeatedly pose queries, interpret responses, reformulate prompts, and verify results. This process closely mirrors the iterative logics described by Bates and Marchionini, positioning berrypicking and exploratory search as precursors to prompt-based information behaviour.

AI-mediated settings offer a distinct empirical advantage because they preserve interaction sequences with a degree of continuity and granularity that is particularly useful for tracing successive reformulations, evaluative moves, and requests for justification. These traces capture queries alongside micro signals of hesitation, uncertainty, trust calibration, and verification, which in many earlier settings required supplementary methods to observe with comparable continuity. Generated naturally and with high temporal resolution, such data provide a trace rich setting for examining mechanisms that have often remained difficult to operationalise within broader models of information behaviour, including barriers, intervening variables, and activating conditions discussed by Wilson (1981). Importantly, using GenAI for information seeking does not reduce these systems to the risky role of simple information sources. The same interactional affordances that produce rich behavioural traces also support reflective use, with GenAI acting as a cognitive intermediary that helps structure problems, articulate needs, compare strategies, and identify external sources. Used with awareness of their limitations, GenAI systems function not as substitutes for publications or databases, but as mediators guiding users through the wider information landscape.

This convergence of dialogic retrieval and mechanistic analysis extends rather than displaces earlier traditions of log-based and process-oriented inquiry in information behaviour research. AI-mediated interaction makes iterative behaviour more readily traceable across successive turns and therefore provides a particularly useful setting for linking Wilson's macro-level categories to candidate mediating mechanisms and observable indicators.

Aim and contribution

The aim of this paper is to develop a conceptual and methodological framework that reconsiders Wilson's model of information behaviour as its central conceptual reference point, while situating Bates's berrypicking and Marchionini's exploratory search as important antecedents for the iterative and dialogic dimensions of information interaction. The argument is developed from a mechanism oriented and falsification sensitive perspective. The present study explores how selected activation and mediation processes associated with Wilson's framework may be reformulated as candidate mediators in dialogic, AI mediated contexts. By combining Popper's and Lakatos's views of scientific demarcation with Cronbach and Meehl's (1955) framework of construct validity, the paper proposes a structured way to translate conceptual variables into measurable indicators.

At the conceptual level, the paper clarifies how Wilson's model may be situated within a broader hierarchy of information behaviour formulations, ranging from heuristic description to more explicitly testable arrangements of constructs and claims. At the methodological level, it introduces the Dialogic-Mechanistic Information Behaviour Framework as a proposal for linking Wilson's macro-level categories with candidate cognitive and affective mediators, such as perceived cost, risk appraisal, trust calibration, and epistemic vigilance, and with possible trace-level indicators in AI-mediated dialogue. More broadly, the paper argues that stronger attention to empirical vulnerability may support more cumulative development in information behaviour theory without discarding the integrative value of classical models.

More specifically, DMIBF is proposed as a conceptual-methodological bridge between Wilson's macro-level categories, candidate cognitive and affective mediators, and possible trace-level indicators in AI-mediated dialogue. Any empirical material discussed later in the paper is used solely for illustrative purposes, in order to show how such mappings and first-level hypotheses may be operationalised, rather than to provide empirical validation of the framework itself. In doing so, the paper adopts a falsification-oriented stance by specifying boundary conditions and observable disconfirmation criteria, while treating the present article as a proposal for more rigorous future testing rather than as a completed empirical test of Wilson's model (Hagger, Gucciardi & Chatzisarantis, 2017; Salovaara & Merikivi, 2015).

For terminological clarity, three terms are used in a distinct sense throughout the paper. Wilson's model is treated primarily as a macro-model, that is, a high-level integrative architecture organising major categories and relations in information behaviour. The DMIBF is presented as a framework in the conceptual-methodological sense, because it structures the translation from macro-level categories to candidate mediators and observational indicators. First-level hypotheses, in turn, refer to propositions linking such candidate mediators to observable behavioural traces. They do not directly test Wilson's macro-model as a whole but are intended to support later empirical examination of broader theoretical claims.

Theoretical Background

Iterative and dialogic LIS models: Bates and Marchionini

Research on information behaviour shows that information seeking is an adaptive, iterative process rather than a linear progression from query to answer. This perspective is established by influential models such as Bates' berrypicking model and Marchionini's exploratory search. Bates (1989) introduced berrypicking to describe how users incrementally "pick" information from heterogeneous sources over time, reformulating queries as tasks and understanding evolve. The model emphasises flexibility and micro-strategies, including query reformulation, citation chaining, serendipitous discovery, and lateral movement across sources and genres. Relevance is treated as dynamic, with information needs progressively respecified during the search, shifting the unit of analysis from a single query to the overall search trajectory. Subsequent work (Bates, 2002) integrated seeking and searching, reinforcing iteration as a core feature of human information interaction.

Marchionini (1995; 2006) developed a complementary account of exploratory search, framing search as a learning-oriented, open-ended activity rather than a quest for a known item. Exploratory search integrates lookup, learning, and investigation, with users asking, examining, navigating, and reflecting. The 1995 monograph conceptualises search as a multi-stage process in electronic environments, while the 2006 article highlights the shift from finding to understanding through metacognitive regulation, opportunistic strategy changes, and tolerance for ambiguity.

Comparing both models reveals three shared claims. First, iteration is intrinsic: queries diversify and are revised as needs evolve. Second, adaptation is central: users select tactics (clarify, broaden, narrow, pivot) in response to feedback. Third, progress depends on dialogic user–system interaction, even in pre-AI contexts. Although Bates and Marchionini studied online catalogues and Web search, their process logic remains highly relevant to dialogic prompting with LLMs: users issue an initial prompt, inspect output, reformulate requests, seek justification or sources, and pursue emerging facets. Both contributions function primarily as heuristic and process-oriented frameworks. They richly describe micro-strategies, phases, and cognitive aims, but they were not primarily formulated as predictive systems designed to yield explicit, falsifiable claims about the occurrence, frequency, or outcomes of specific tactics. Concepts such as exploration, sense-making, and serendipity are analytically valuable yet underspecified as testable variables, a limitation repeatedly noted in the literature (e.g., Case & Given, 2016; Savolainen, 2007; Julien & Duggan, 2000). In sum, Bates and Marchionini offer a procedural vocabulary for iteration and adaptation that is valuable for analysis and design, but less suited to direct falsification at the level of formally specified predictions. This characterisation aligns with later syntheses arguing that canonical information-seeking models operate as heuristic maps rather than empirically refutable predictive systems (e.g., Salovaara & Merikivi, 2015).

LLM-mediated conversational retrieval renders dialogic interaction explicit: response histories condition subsequent outputs, and users iteratively refine prompts in relation to considering content, style, evidence, or perceived hallucinations. The feedback-sensitive moves described by Bates and Marchionini are now captured at fine temporal resolution in prompt–response logs, making certain interactional micro-dynamics more directly observable than in many earlier settings. This observational advantage does not replace earlier process-oriented traditions in information behaviour research; rather, it extends them by providing a trace-rich setting in which iterative behaviour can be examined with greater continuity across successive turns.

Wilson’s macro-model and its epistemic limits

Tom D. Wilson developed a family of models introduced in 1981, 1996, and 1999 that broaden the scope from tactical iteration to the conditions under which information seeking is initiated, sustained, and used. These models are widely cited as macro-frameworks because they integrate cognitive, affective, social, and environmental dimensions into a single architecture.

Wilson (1981) articulated an early scheme linking information need (rooted in role-related and psychological contexts) to information-seeking behaviour and information use. The emphasis is on why people seek information (e.g., to resolve uncertainties, support decisions) and how contextual factors shape the possibility and form of seeking. The 1996 model consolidated and extended this account by introducing intervening variables such as psychological, demographic, role-related, interpersonal, and environmental factors that affect whether and how needs lead to seeking. Crucially, Wilson distinguishes barriers (e.g., access, time, anxiety) from motivational components. Subsequently, Wilson (1999) refined the model by highlighting activating mechanisms (e.g. stress/coping, risk/reward), thereby clarifying how needs are translated into behaviour given intervening variables. In that paper, Wilson presents the model primarily as an organising device for research and for the integration of findings across studies.

Four components are central across these iterations: (i) information need, (ii) intervening variables, (iii) barriers (a subset with negative valence), and (iv) activating mechanisms that trigger or inhibit seeking and subsequent use. As a macro-model, Wilson's framework aims for coverage and integrative power. Its strength lies in providing a high-level explanatory architecture that accommodates contextual, motivational and behavioural dimensions within a single scheme.

Epistemic limits. The strength of Wilson’s approach, its breadth and integrative capacity, also marks its methodological challenge. Constructs such as intervening variables and activating mechanisms are analytically rich but empirically underspecified: the framework names families of influences without stipulating testable functional forms (e.g., direction, magnitude, thresholds) that could be subjected to disconfirmation. Wilson himself characterises the model as an analytical rather than a predictive system (Wilson, 1999). Subsequent reviews observe that many information-behaviour models (including Wilson’s) function as conceptual maps more than empirically refutable theories (e.g., Case & Given, 2016; Julien & Duggan, 2000; Savolainen, 2007). The broader literature similarly treats Wilson’s framework, alongside other canonical models such as Dervin's, Ellis's and Kuhlthau's, as largely descriptive, valuable for organising findings but rarely framed to risk empirical failure (Salazar et. al., 2007; Zeng, 2011). The practical upshot is a validation gap: it can be difficult to derive hypotheses that would put the model at risk of empirical rejection rather than merely accommodate observed patterns post hoc.

We therefore refer to Wilson's contribution as a macro-model, a high-level architecture that enumerates components and relationships. At the same time, its integrative scope gives it some meta-model-like features, insofar as it can accommodate more local accounts of behaviour within a broader explanatory structure. For present purposes, the macro-model label is retained because it better captures the paper's present concern with high-level explanatory architecture rather than with a formal model of models. A falsification-oriented programme therefore requires explicit specification of disconfirming observations and testable boundary conditions, rather than further integrative restatements (e.g. Gamayunov, 2014).

Locating the operational challenge. If intervening variables and activating mechanisms cannot be measured directly in a stable way across contexts, then hypothesis testing risks becoming underdetermined. The field needs principled procedures for: (a) mapping Wilson's conceptual entities onto observable indicators; (b) specifying directional claims about their relations; and (c) identifying boundary conditions (e.g., task complexity, domain familiarity) under which predictions should or should not hold.

The Dialogic-Mechanistic Information Behaviour Framework

The Dialogic-Mechanistic Information Behaviour Framework (DMIBF) extends Wilson's macro-model by embedding it within a multi-level architecture that translates conceptual constructs into more explicitly operationalisable processes. It articulates three interconnected layers: conceptual, mechanistic, and observational, each corresponding to a distinct level of theoretical abstraction and empirical access. This layered design is intended to support the formulation of more explicit and potentially testable propositions in AI-mediated information behaviour, rather than to claim empirical validation in itself.

The conceptual level: Wilson's macro-variables

At its highest level, the DMIBF retains the theoretical structure of Wilson's models (1981, 1996, 1999). These macro-variables represent the foundational architecture of information behaviour:

- information need - the cognitive recognition of a knowledge gap;
- intervening variables - psychological, social, and environmental factors mediating action;
- barriers - constraints (situational, cognitive, affective) impeding information seeking;
- activating mechanisms - motivational triggers that transform latent need into observable behaviour;
- information seeking, processing, and use - the external manifestations of behaviour.

In Wilson's formulation, these components were explanatory categories rather than measurable entities. Their relationships were specified at a descriptive level, generating broad theoretical coherence but lacking operational precision. Within the DMIBF, these same categories are retained as conceptual priors, or, in Lakatosian terms, as elements of a relatively stable theoretical core, while being linked more explicitly to lower-level mechanisms and indicators.

The mechanistic level: cognitive and affective mediators

The second layer explains how macro-variables generate behaviour via cognitive and affective mediators, providing a mechanistic translation of Wilson's intervening variables and activating mechanisms. Cognitive mediators regulate evaluation and refinement across dialogic turns, shaping prompt precision and verification. Affective mediators modulate motivation and persistence in information interaction.

These mechanisms can be conceptualised as entities and activities organised to generate behaviour (Illari & Williamson, 2012; Machamer, Darden & Craver, 2000). For instance, a rise in epistemic vigilance (a metacognitive activity) should systematically increase the number of verification prompts (an observable behaviour). Conversely, elevated cognitive cost (an affective mediator) should reduce iteration frequency. In both cases, the mechanism entails counterfactual commitments: if the mediator is altered, the behavioural outcome should predictably change, thereby creating conditions for more explicit empirical vulnerability.

This layer thus performs a mechanistic decomposition of Wilson's model, bridging the conceptual and empirical realms. Where the original model posited static "intervening variables", the DMIBF identifies dynamic processes that can be traced, measured, and modelled.

The observational level: behavioural traces in AI-mediated interaction

At the lowest level, the DMIBF anchors mechanisms in observable behavioural traces generated during dialogic interaction with AI systems. These traces form what Craver and Darden (2013) call behavioural mechanism evidence: concrete patterns through which mechanisms manifest in real time.

Key observational indicators include:

- prompt formulation and reformulation rate (indexing metacognitive awareness);
- hedging and uncertainty markers (indexing affective uncertainty);
- verification prompts and justification requests (indexing epistemic vigilance);
- iteration depth and session length (indexing motivation and perceived cost);
- evaluative and integrative turns summaries, paraphrases, or applications (indexing information use and reflection).

These data are natively recorded in AI dialogue logs, providing high-resolution temporal sequences suitable for quantitative and qualitative analysis. Because the same indicators can be aggregated or compared across users, tasks, or systems, they support progressive refinement in the Lakatosian sense, new hypotheses can be tested without altering the theoretical core.

Architectural coherence and theoretical implications

The DMIBF employs a structured architectural design, wherein each level imposes constraints and provides validation for the succeeding levels indicated in the Tab. 1.

Level	Function	Example
conceptual (Wilson)	defines theoretical structure and relations among variables	information need → intervening variables → behaviour
mechanistic (cognitive-affective)	specifies causal processes and mediators	epistemic vigilance, trust calibration, uncertainty management
observational (dialogic data)	provides measurable indicators	verification prompts, reformulations, hedging markers

Table 1. Conceptual, mechanistic and observational levels in the DMIB Framework; source: self-authored (2026)

This tri-level design supports a recursive research cycle:

1. Theory articulation (conceptual);
2. Mechanism specification (mechanistic);
3. Empirical testing and falsification (observational).

When data support directional predictions, the programme may be regarded as progressive in Lakatosian terms; when they fail, mechanisms can be revised without abandoning the conceptual framework. In this sense, the DMIBF offers a way of operationalising Lakatos's vision of a progressive research programme in information behaviour, one that grows through empirical challenge rather than conceptual accretion.

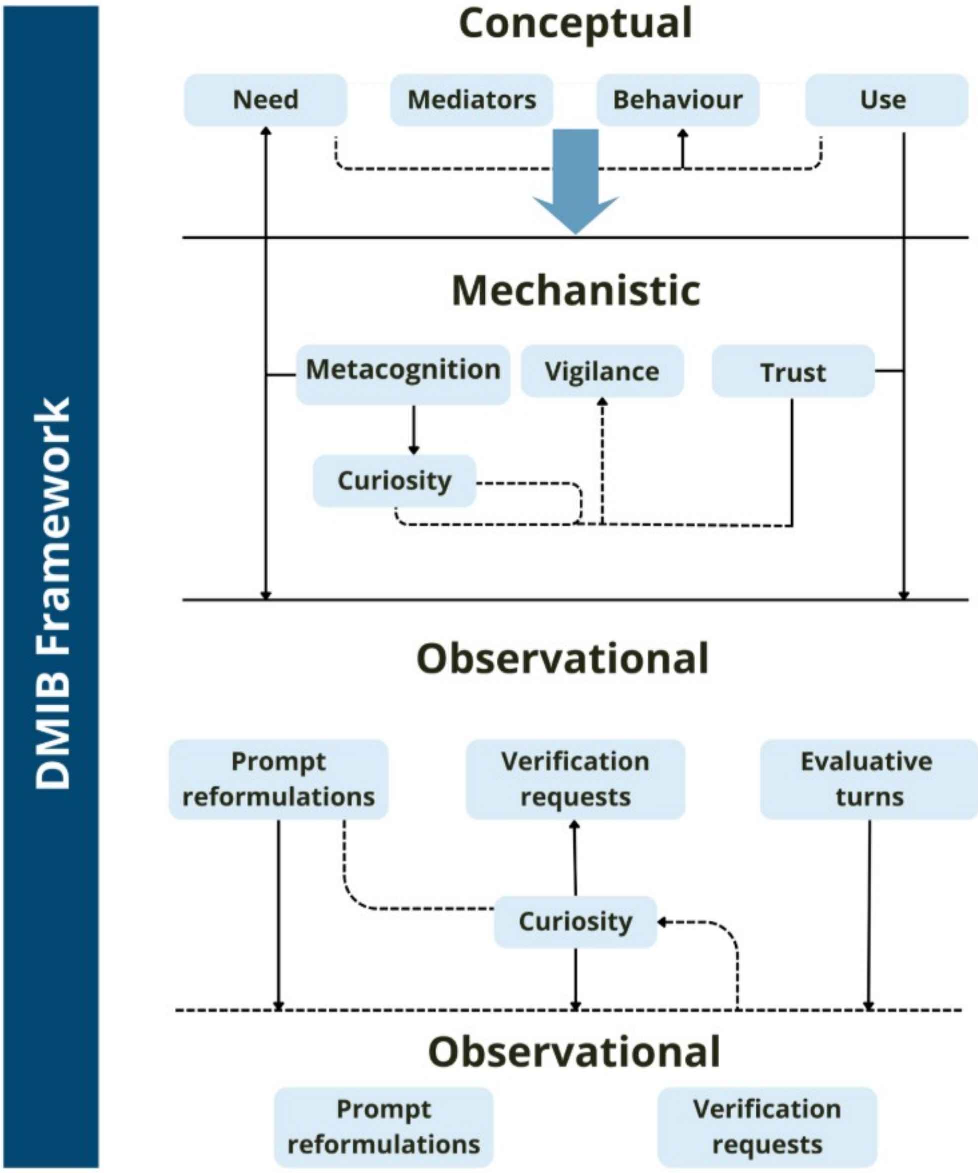

Figure 1. Diagrammatic representation of Dialogic-Mechanistic Information Behaviour Framework (DMIBF); source: self-authored (2026)

The DMIBF is both explanatory and methodological, grounding Wilson's macro-model in mechanistic specificity and greater empirical tractability. Its tri-level architecture links theory, mediators, and observable data, while drawing on Popper's criterion of falsifiability, Lakatos's requirement of theoretical progress, and Cronbach & Meehl's standard of construct validity. Accordingly, AI-mediated dialogue is treated here as a particularly useful trace-rich setting for the operationalisation and future empirical examination of information-behaviour mechanisms, rather than as a self-sufficient source of validation. In this sense, the framework should be understood as providing a structured pathway for operationalisation and future empirical testing, rather than as constituting empirical validation in itself.

Theoretical and methodological grounding of the DMIBF

From metaphor to mechanism: the mechanistic frame

Moving from descriptive adequacy to empirical testability requires specifying mechanisms—entities and activities organised to produce the phenomenon of interest. In the philosophy of science, Machamer, Darden, and Craver (2000) formalised this view, arguing that mechanistic explanations identify components and operations and show how their organised interaction generates behaviour. Later work extends this account by distinguishing how-possibly and how-actually models and by emphasising the constraints that prevent mechanistic explanation from collapsing into post hoc narrative (Craver & Darden, 2024; Craver, Tabery & Illari, 2015). Although developed in biology and neuroscience, this framework has informed social-scientific modelling by decomposing complex systems into interacting processes.

Mechanistic specification offers two epistemic benefits that address previously noted limitations (cf. Savolainen, 2007; Case & Given, 2016). First, it enables operationalisation by linking theoretical entities to observable activities, thereby providing a concrete measurement strategy. For example, if epistemic vigilance is treated as a mechanism governing claim acceptance, its operation can be provisionally indexed through behavioural traces such as verification prompts or requests for sources. Second, mechanistic specification secures falsifiability: mechanisms entail counterfactual commitments, such that perturbing a component should yield predictable changes in outcomes. These commitments translate theory into testable hypotheses with directional predictions and, ideally, threshold or interaction terms that render models empirically vulnerable to disconfirmation. In the present paper, this mechanistic perspective is used not as an empirical proof in itself, but as a conceptual-methodological means of clarifying how Wilson's broad categories may be translated into more explicitly testable propositions.

Reinterpreting Wilson mechanistically

The relevance of mechanistic reasoning becomes evident when applied to established frameworks in information behaviour, particularly Wilson's macro-model. While the model integrates information needs, context, mechanisms, and behavioural outcomes, it does not specify their organised causal interactions. A mechanistic reinterpretation thus treats the model as a scaffold of regulated transitions rather than primarily as a descriptive map, with components linked to candidate entities or activities observable in AI-mediated interaction (Wilson, 1999).

Information need → initial condition (user-side input).

From a mechanistic perspective, the initial condition is consequential because it shapes downstream trajectories of interaction. In AI-mediated contexts, one observable manifestation of this initial condition is the articulation of the need in the opening prompt, including task constraints, domain terminology and explicit success criteria. Importantly, the relationship between this articulation and output quality is moderated by system-level conditions such as model version, tool availability (e.g. external search or citation functions) and generation parameters. These conditions are not part of the construct of information need but constitute boundary conditions that must be reported to render subsequent claims falsifiable.

Intervening variables → cognitive and affective mediators.

Wilson's intervening variables can be reinterpreted mechanistically as internal regulators shaping how articulated needs translate into action. In this framework, mediators such as metacognitive awareness, perceived cognitive cost, uncertainty management, trust calibration, and epistemic vigilance are linked to trace indicators in dialogic logs (e.g., self-corrections, hedging, verification prompts). These indicators are treated as provisional proxies rather than direct measures of internal states and require validation through first-level hypotheses.

Activating mechanisms → motivational processes.

Wilson's activating mechanisms can be conceptualised as motivational processes regulating the balance between exploration and evaluation. Curiosity may drive exploratory branching, while risk–reward appraisals shape preferences for verification versus speed. In dialogic traces, these processes appear as shifts between exploratory and evaluative intents, requests for justification, or comparisons of alternatives.

Behaviour and use → output-side activities.

Behavioural outcomes and information use can be operationalised as observable interactional activities in the prompt stream. Indicators include iteration count, reformulation rate, intent diversity, explicit evaluation moves, and integration behaviours such as summarising or applying outputs. While not mechanistic in themselves, these activities provide the observational layer for testing mechanistic hypotheses and for examining how proposed mediator-indicator relations may be specified under particular task and system conditions.

Taken together, this reinterpretation shifts the model from a metaphor of influence, in which variables intervene, to a mechanism-oriented account in which mediators regulate transitions from need to action under specified boundary conditions, leaving observable correlates in AI-mediated dialogue. In this sense, the mechanistic reinterpretation does not yet validate Wilson's model empirically; rather, it provides the conceptual and methodological bridge required for more explicit operationalisation and subsequent testing.

AI-mediated environments: theoretical rationale

Dialogic interaction with LLMs offers confers three methodological advantages that render mechanistic approaches practical. (1) Granular traces: each prompt and response is recorded as time-ordered text. Self-corrections, hedges ("maybe…", "I think…"), verification requests ("give sources", "justify") and stance shifts are visible. These traces provide possible behavioural proxies for internal mediators (e.g., uncertainty management). (2) Iterative cycles: the unit of analysis is a multi-turn session, which permits closer observation of within-session change (e.g., does perceived cost reduce exploration as sessions progress?). (3) Prompting and task structures can often be standardised, allowing variation in task properties, such as complexity, stakes, and domain familiarity. This approach facilitates exogenous variation, which is crucial for the testing of boundary conditions and the validation of hypotheses. These advantages should not be understood as wholly replacing earlier approaches to information behaviour research; rather, they provide a setting in which iterative and evaluative processes can often be observed with greater continuity and traceability across successive turns.

From mechanisms to testable propositions

To be falsifiable, Wilson-based claims must leave the safety of post hoc accommodation. Mechanistic commitments allow one to derive directional hypotheses linking mediators to behavioural indicators, for example:

- higher metacognitive awareness (mediator) → greater initial prompt precision and fewer early reformulations (behavioural indicators).
- lower perceived cognitive cost combined with calibrated trust (mediators) → higher iteration rates and deeper dialogic engagement.
- greater affective uncertainty (mediator) → lower prompt precision and higher reliance on confirmation requests.
- stronger epistemic vigilance (mediator) → more frequent verification prompts and higher factual accuracy in accepted outputs (assessed, for example, by independent coding against a reference set or by source-based verification).

- more reflective evaluation (mediator) → fewer uncritical acceptances of erroneous content and greater evidence of learning transfer (summarising, paraphrasing, application).

These claims do not restate the theories of Bates or Marchionini but translate Wilson's components into mechanistic, observable terms within AI-mediated interaction. At this stage, however, they should be understood as framework-generated propositions rather than as hypotheses already examined within the present paper. Crucially, each claim is open to refutation; for example, if increased vigilance does not predict verification behaviour or accuracy, the claim fails in that context.

A key source of confusion is that translating Wilson's macro variables into falsifiable propositions involves two analytically distinct steps within a single methodological pipeline. The first is operational and concerns construct validity: Wilsonian categories are mapped onto candidate mediators and linked to observable dialogue indicators, a step not specified in the original model and therefore requiring explicit justification. The second is mechanistic and concerns empirical vulnerability: once this mapping is sufficiently justified, directional, counterfactual hypotheses specify how changes in mediators should affect behaviour. First-level hypotheses concern the plausibility and coherence of mediator-indicator links, while higher-level hypotheses test the broader Wilsonian architecture. This distinction clarifies that falsifiability rests on pre-specified mappings, predictions, and scope conditions, rather than post hoc interpretation. In the present paper, the emphasis falls primarily on the first of these steps: the conceptual-methodological specification of mediator-indicator mappings and the formulation of candidate first-level hypotheses that may support later empirical testing of broader Wilsonian claims.

Caveats and scope conditions

Two cautions apply when treating behavioural traces as evidence of mechanisms. First, proxies imperfectly map onto internal states: hedging may signal epistemic uncertainty but can also reflect politeness norms or cultural interaction styles. Second, verification actions may be instruction-driven rather than indicative of epistemic vigilance. Disentangling these possibilities requires larger or longitudinal interaction corpora that allow comparison of patterns across users, tasks, and contexts. If verification proves situational rather than dispositional, vigilance-based hypotheses must be rejected under that boundary condition.

Mixed-methods designs combining interaction traces with concurrent think-aloud protocols and post-task reports can reduce interpretative underdetermination and the risk of misattributing behavioural markers to inappropriate cognitive or affective mechanisms (e.g., Boren & Ramey, 2000; Fox, Ericsson & Best, 2011). System dynamics also matter: LLM outputs are stochastic and instruction-tuneable, and identical prompts can elicit different behaviours across systems or versions. Accordingly, hypotheses must explicitly specify the system and task context (e.g., analytical versus creative tasks).

Within a mechanistic perspective, Bates and Marchionini retain their status as process-level descriptors of iteration and exploration, now embedded in a causal system that explains when and why these processes occur. Wilson contributes a high-level structural framework, while mechanistic reinterpretation supplies the operational links needed to integrate these elements. In this sense, AI-mediated environments should be treated as a particularly useful empirical setting for examining such links, rather than as a complete methodological break with earlier traditions of information behaviour research (Case & Given, 2016; Julien & Duggan, 2000; Savolainen, 2007; Wilson, 1999).

Methodological orientation

The present study adopts a conceptual-methodological orientation designed to clarify how the Dialogic-Mechanistic Information Behaviour Framework may support more explicit operationalisation of Wilson's model. Rather than introducing a separate empirical study at this stage, the section specifies the epistemological and methodological principles that guide the translation of macro-level constructs into candidate mediators, observable indicators, and first-level hypotheses. In this sense, the methodological task is not to validate the framework empirically within the present paper, but to define the inferential conditions under which such validation may later be pursued.

Conceptual-methodological synthesis

The methodological logic of the paper draws selectively on three complementary traditions: Popper's criterion of empirical vulnerability, Lakatos's distinction between theoretical core and auxiliary refinement, and Cronbach and Meehl's account of construct validity. Together, these perspectives provide a basis for specifying how broad conceptual categories may be linked to observable indicators without collapsing theory into post hoc description. Within the present argument, Popper clarifies the importance of disconfirming observations, Lakatos helps distinguish relatively stable conceptual commitments from revisable operational mappings, and Cronbach and Meehl provide the methodological rationale for linking constructs to indicators within a defensible nomological network (Cronbach & Meehl, 1955; Hagger, Gucciardi & Chatzisarantis, 2017; Lakatos, 1970; Popper, 1959; 2014).

Contemporary work on mechanisms and validity further supports this orientation. Mechanistic explanation clarifies how candidate processes may connect conceptual categories with observable activity, while later developments in validity theory stress that inferential adequacy depends on the coherence of the link between theory and evidence rather than on measurement alone (Craver & Darden, 2013; Borsboom, Mellenbergh & van Heerden, 2004; Illari & Williamson, 2012; Kane, 2013). Taken together, these traditions justify the present focus on mediator-indicator mappings, scope conditions, and explicitly formulated first-level hypotheses.

From constructs to indicators and first-level hypotheses

This section sets out how Wilson's macro-level categories are translated into candidate mediators and observable indicators within the DMIBF. Each proposed relation is specified in terms of a construct-to-indicator mapping located within a nomological network and in terms of disconfirming observations that would count against that mapping. Scope conditions, including task type, domain familiarity, and system configuration, are stated explicitly, in line with the view that theoretical progress depends on boundary testing rather than on the accumulation of confirmations (Ajzen & Fishbein, 2004; Gamayunov, 2014; Hagger, Gucciardi & Chatzisarantis, 2017; Salovaara & Merikivi, 2015).

No.	Wilson's Construct	Cognitive / Affective Mediator (Mechanism)	Observable Indicator (AI-mediated environment)
1	**Information need** *trust calibration* (assessment of the AI's reliability and cognitive cost of iteration)	*Metacognitive awareness* and *problem articulation* (the user's self-reflective recognition and expression of a knowledge gap)	Observable indicators: initial prompt specificity, early reformulations, self-correction markers, metacommunicative statements.
2	**Motivation / Cost-Benefit Evaluation**	*Perceived cognitive cost* and *trust calibration* (assessment of the effort of iteration and the provisional reliability of AI output)	Observable indicators: iteration count, time on task, trust or doubt markers, delegation requests, verification intensity.
3	**Barriers (psychological, situational, or contextual)**	*Uncertainty management* and *affective regulation* (ability to handle ambiguity, frustration, or uncertainty in AI responses)	Observable indicators: hedging and clarification, frustration markers, tone shifts, reset moves, escalation after failed cycles.
4	**Activating mechanisms**	*Curiosity* and *epistemic vigilance* (motivation to verify or refine AI-generated information)	Observable indicators: intent diversity, branching and side threads, verification prompts, counterargument requests, perspective switching.
5	**Information processing and use**	*Reflective evaluation* and *learning transfer* (active integration of AI outputs into user reasoning)	Observable indicators: evaluative turns, paraphrasing and summarising, inconsistency detection, source comparison, learning transfer statements.

Table 2: Links between Wilson's macro-variables and operational indicators; source: self-authored (2026)

Table 2 summarises how Wilson's macro-level constructs may be linked to cognitive and affective mediators and, in turn, to trace-level indicators in AI-mediated interaction. At this stage, the table should be understood as an instrument of conceptual-methodological specification rather than as evidence that such mappings have already been empirically confirmed. Its role is to make explicit the inferential pathway from theoretical category to candidate indicator and thereby to support later empirical examination under stated boundary conditions. The full version is provided in Appendix 1.

The framework treats Wilson's constructs as candidate cognitive and affective mediators, thereby forming a mechanistic bridge to observable behaviour. These mediators are informed by established traditions of psychological and behavioural research, while the observational layer captures trace-level patterns in GenAI interaction as provisional evidence of mechanism activation. Consequently, Table 1 offers an operational scaffold for more explicit empirical examination, rather than empirical confirmation in itself.

First-level hypotheses

First-level hypotheses link candidate mediators to observable behaviour in GenAI interaction. They do not directly test Wilson's macro-model as a whole; rather, they examine whether proposed mediator-indicator mappings are sufficiently coherent and explicit to support later testing at a broader theoretical level. In this sense, first-level hypotheses function as an intermediate layer between conceptual reconstruction and future empirical evaluation. At a general epistemological level, theory appraisal may involve both empirical disconfirmation and logical inconsistency; however, the present section is concerned primarily with empirically assessable mediator-indicator relations. Multiple hypotheses may be specified for a given mediator, with their selection shaped by the scope and design of a later empirical study. A full list of first-level hypotheses developed in this project is provided in Appendix 2; the following paragraphs present only selected illustrative examples of first-level hypotheses for specific constructs.

H1. 1.2. Users with higher metacognitive awareness more often use metacommunicative messages about their own state of knowledge, e.g., "I'm not sure," "I think I misunderstand."

H1. 2.1. Users with higher levels of trust are more likely to express explicit signals of acceptance ("okay, sounds good," "I trust that this is correct").

H1. 3.3. Users with high uncertainty management use coping strategies (clarification, reframing the task) and do not escalate even after a series of AI errors.

H1. 4.2. Users with high epistemic vigilance are more likely to use explicit verification prompts ("provide sources," "prove it," "is this consistent with the literature?").

H1. 5.2. More frequent reformulation of AI responses in one's own words ("so what you're saying is...") indicates a high level of reflective information processing.

The formulation of first-level hypotheses should consider whether users engage with GenAI as a tool or as a conversational partner; when interaction style is not pre-selected, both variants should be operationalised. The illustrative material discussed below draws on student–GenAI interaction transcripts (<anonymized for review purposes>) and is supported by an illustrative codebook (Appendix 3) with six top-level codes aligned with Wilson's constructs (Table 1).

Illustrative operationalisation of selected first-level hypotheses

The brief examples below are included solely to illustrate how selected first-level hypotheses may be interpreted in context. They do not constitute a systematic empirical test of the framework and should not be read as preliminary validation.

To illustrate H1.1.1., opening prompts such as "Do you know the Academy of Fine Arts (ASP) in Krakow?" (Case C, Item 20) and "Are you familiar with the concept of transferring human consciousness to a computer?" (<anonymized for review purposes>, Item 20) require careful interpretation. While they may reflect anthropomorphism and limited trust in AI capabilities, they can also function strategically to initiate quasi-human interaction. Taken in isolation, they reveal little about AI literacy, with their significance emerging only in longer interaction sequences.

Metacommunications about the state of knowledge (H1.1.2), such as "Did you understand the command?" or competence-checking prompts, signal both an attempt to establish mutual understanding and the user's own knowledge state. While similar to common-ground strategies in human communication, in anthropomorphised GenAI contexts they may also reflect uncertainty, making their interpretation dependent on the broader interactional context.

In signals of doubt (H1.2.2), users may question either information reliability or the problem situation itself. Questions such as “Do you think that such a book has any chance of success on the current book market…?” (<anonymized for review purposes>, Item 111) direct uncertainty to the AI and, following Dan Sperber and Deirdre Wilson’s (1996) relevance theory, presuppose that delegating the question to the system may help reduce uncertainty, reflecting either trust or a testing strategy. By contrast, doubts about omissions or response quality—for example, “You didn't mention anything about Alien: Isolation…” (<anonymized for review purposes>, Item 51)—presuppose evaluative capacity and may indicate higher AI literacy.

Changes in user-GenAI communication may appear as shorter, less detailed prompts (cf. H1.2.4, H1.2.5, H1.4.1). Such simplification is ambiguous, potentially indicating either declining engagement or task completion, and therefore requires interpretation within the broader interactional context rather than relying on prompt length alone. Interpretation of these changes is supported by explicit user evaluations of the output (e.g., satisfaction or quality assessments; see H1.2.1) and by subsequent uses of AI-generated information (see H1.5.2, H1.5.3), such as paraphrasing, integration into new queries, or transfer to later tasks. In these cases, shorter prompts may indicate task completion rather than reduced engagement (see H1.4.2).

Scope conditions and interpretive limits

Failed predictions are treated as informative constraints rather than anomalies. At the first-level mapping stage, a hypothesis fails when a proposed mediator does not co-vary with its behavioural indicators as predicted; for example, if increased epistemic vigilance does not lead to more verification prompts across tasks or after questionable outputs, the mediator–indicator mapping is disconfirmed. At the macro level, failure concerns whether mediators regulate transitions from need to behaviour and use. When predicted relations (e.g., between cost, trust, iteration, or verification and acceptance quality) do not hold, mechanisms or scope conditions are revised rather than the framework rejected, with disconfirmations serving to sharpen theoretical scope.

Within the DMIBF, falsifiability is approached through the explicit specification of links between conceptual categories, candidate mediators, and observational indicators. These links permit the formulation of directional propositions that may later be subjected to empirical challenge. At the present stage, however, the framework should be understood as providing the conceptual and methodological conditions for such testing, rather than as establishing a predictive system already validated by the material discussed in this paper.

Illustrative vignettes: trace evidence with participant-provided context

In the illustrative corpus used here, interactions with GenAI average 4.6 prompts, indicating relatively brief exchanges that may limit observation of diverse strategies. However, user-written reflections on task motivation and outcome assessment complement the interaction logs, providing important contextual insight into users’ intentions and evaluations of the generated results.

The illustrative vignettes below are intended to show how participant-provided context may assist in interpreting trace indicators that would otherwise remain underdetermined. They are not presented as a systematic dataset for testing the framework, but as examples of how operationalisation may be contextually constrained.

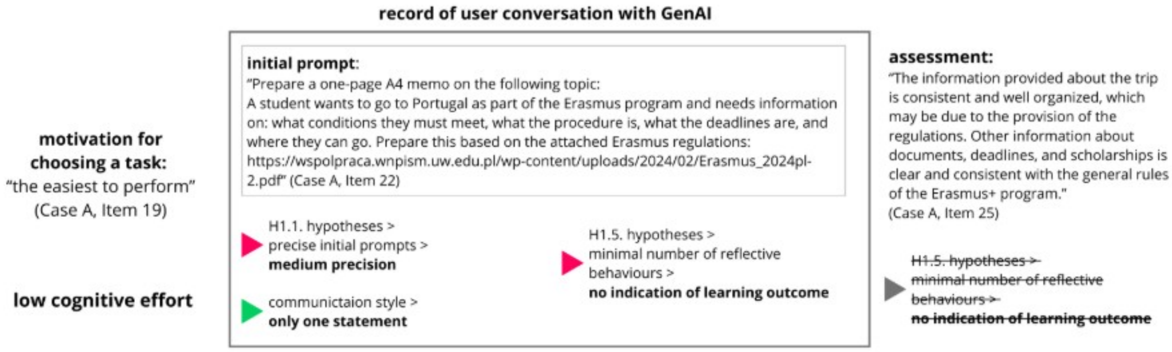

Figure 2. Vignette A: one-turn interaction and early closure; source: self-authored (2026)

The first vignette (Fig. 2) shows an interaction ending after a single prompt–response exchange. While trace data alone are ambiguous, the participant's report of low task difficulty indicates satisficing under low perceived cost, explaining the absence of iterative or evaluative behaviour. The case underscores the need to specify boundary conditions when interpreting first-level hypotheses.

The second vignette (Fig. 3) illustrates sparse prompting combined with extensive post-task evaluation, indicating trust calibration and epistemic vigilance through selective acceptance and rejection of unreliable output. This shows that verification can occur via reflective assessment, not only through explicit prompts. The full participant assessment is provided in Appendix 4.

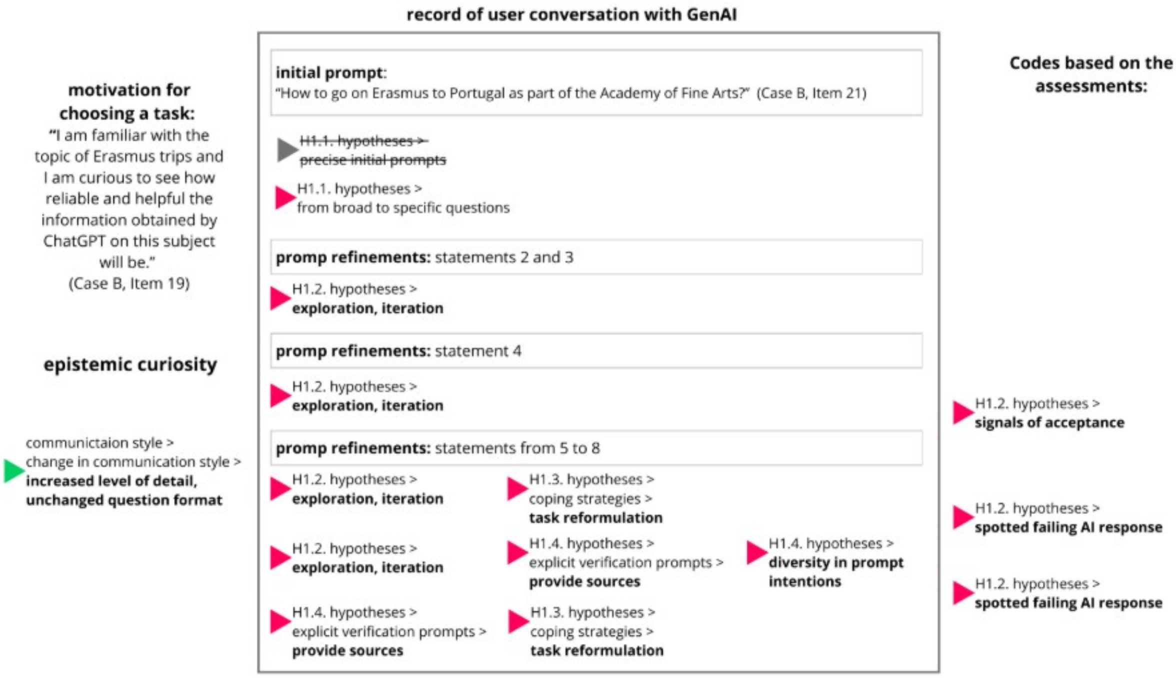

Figure 3. Vignette B: economical prompting with detailed reflective evaluation; self-authored (2026)

The third illustration (Fig. 4) shows a long interaction that begins with an open, knowledge-oriented prompt and develops into increasingly detailed and critical engagement. Motivated by interest and low trust in the AI, the user tests responses, identifies inaccuracies, and paraphrases outputs to verify them, evaluating results against their own (partly incorrect) prior knowledge.

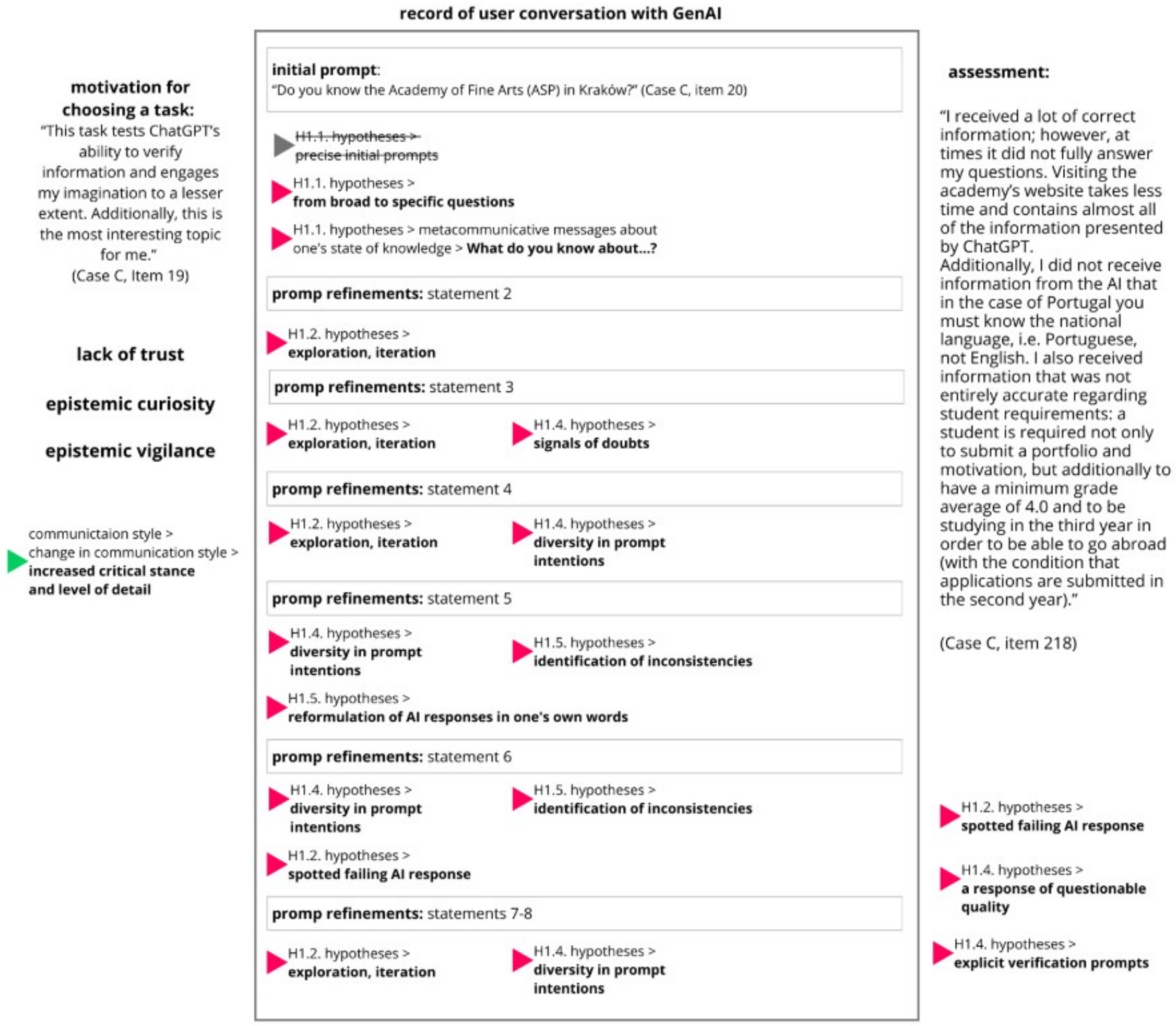

Figure 4. Vignette C: from broad to specific questions while testing GenAI, source: self-authored (2026)

Discussion

This paper addresses a longstanding methodological limitation in information behaviour research: influential models often achieve descriptive adequacy without sufficient empirical risk due to under-specified constructs that allow post hoc accommodation. The Dialogic Mechanistic Information Behaviour Framework responds by treating Wilson's macro-model as a conceptual architecture that may become more explicitly open to empirical testing when its categories are translated into candidate mechanistic mediators and related to observable traces of AI-mediated dialogue.

At the conceptual level, the DMIBF clarifies an emerging setting in which seeking, evaluation, reformulation, and use are increasingly interleaved. AI-mediated, multi-turn prompting is therefore best understood as a form of dialogic information engagement rather than as a simple extension of query-based retrieval or as a wholly novel behaviour. This interpretation remains consistent with the procedural emphasis on iteration and adaptation in Bates's berrypicking and Marchionini's exploratory search. What differs methodologically is that AI-mediated interaction records these iterative adjustments at the level of successive turns, thereby making within-session dynamics more directly available for analysis. In this sense, AI-mediated environments provide a trace-rich setting in which otherwise less visible transitions in information interaction may be examined with greater temporal continuity.

Structurally, the framework distinguishes macro-level categories from mid-level causal regulators, thereby clarifying the analytical distinction between high-level explanation and operational specification. While Wilson's model offers an integrative vocabulary, its broad constructs hinder cumulative testing. The DMIBF preserves these constructs conceptually but requires empirical work to pass through a mechanistic layer, with first-level hypotheses testing the link between Wilson's categories and observable behaviour. This distinction reframes failure: when a directional hypothesis fails, the target of revision is the mechanistic mapping or scope conditions, not a global rejection of Wilson's framework. This supports a Lakatosian view of progress, in which the conceptual core remains stable while mechanistic hypotheses become progressively more constrained and empirically specified, provided that their scope conditions and inferential limits are explicitly stated.

The methodological significance of the framework lies in showing how falsifiability may be approached through directional, counterfactual relations between candidate mediators and trace-level indicators. Because behavioural traces are treated as fallible proxies rather than direct indicators, they require triangulation and negative tests to address construct validity risks, such as politeness-driven hedging or instruction-induced verification. Accordingly, the framework emphasises tests that distinguish dispositional tendencies from situational norms, for example by examining stability across tasks or selective responses to questionable outputs.

These caveats underscore the role of AI literacy in the DMIBF. Rather than an external add-on, AI literacy is treated as a contextual factor shaping how mediators manifest and how behavioural indicators are interpreted. Anthropomorphic prompts or challenges may reflect either limited competence, epistemic vigilance, or learned norms; accordingly, AI literacy is modelled as both a background characteristic and an emergent, session-level phenomenon, informed by user reflections beyond interaction logs. Beyond epistemology and method, the framework also has ethical and infrastructural implications. The use of AI dialogue logs in theory-oriented research raises issues of privacy, consent, re-identification, and reproducibility, particularly as AI systems, interfaces, and model versions continue to change. Accordingly, system parameters, prompts, and sampling constraints must be reported with experimental-level rigour to avoid conflating system artefacts with human behaviour.

Finally, the agenda points to future research directions: using the draft codebook to support shared annotation, adopting a staged strategy from validating mechanistic mappings to testing macro-model dynamics, extending observation beyond text in multimodal LLM settings, and employing longitudinal designs to separate stable traits from contextual strategies. Together, these steps outline a cumulative research programme through which the testability of Wilson-informed claims may be strengthened while preserving the integrative value of Wilson's model.

Conclusions

This paper revisits Tom D. Wilson's model of information behaviour to show how a classical macro framework can be reformulated so as to become more explicitly open to empirical testing in AI-mediated, dialogic settings. The claim is not that earlier models are obsolete, but that their testability is limited by operational indeterminacy. AI-mediated interaction helps address this challenge by providing fine-grained traces of seeking, evaluation, reformulation, and use, thereby enabling a more explicit alignment between conceptual categories and observable transitions.

The Dialogic Mechanistic Information Behaviour Framework (DMIBF) provides this pathway by retaining Wilson's macro variables at a conceptual level and embedding them in a tri-level architecture linking them to mechanistic mediators and observable indicators. Within this framework, empirical vulnerability is approached through explicit counterfactual commitments. When predictions fail to hold, the revision target is the proposed mapping or its scope conditions rather than a post hoc accommodation of the framework itself. First-level hypotheses function as intermediate tests of mediator–indicator links, supporting cumulative development across tasks, populations, and systems. Earlier process models remain relevant in this account: Bates's berrypicking and Marchionini's exploratory search describe iterative dynamics that become more explicitly open to mechanistic specification when situated within Wilson's broader framework. AI-mediated prompting makes these dynamics more readily traceable across successive turns, thereby supporting the more explicit formulation of testable claims about when and why iteration occurs or is suppressed.

Several limitations constrain this contribution: behavioural proxies are imperfect, trace data are shaped by politeness, instruction, and system constraints, and LLM outputs are stochastic and version dependent. Accordingly, hypotheses must specify system and task contexts and be tested through triangulation and replication. Within these limits, the DMIBF reframes AI-mediated dialogue as a trace-rich empirical setting in which mechanistic propositions may be more explicitly specified and challenged through disconfirming evidence, while preserving the integrative value of Wilson's model.

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